

Luke Cullen

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science - Minor in Data Science - 3.72 GPA

Orlando, FL

August 2023 – May 2027

EXPERIENCE

Web Developer Intern

October 2025 – Present

CDL Techrangers

Orlando, FL

- Developing web software with the **CDL Techrangers**, a team inside UCF which develops **open-source educational software** used by **100+ institutions** and **tens of thousands of students** worldwide.
- Deployed an open-source **Flask** tool for processing quiz accessibility requests on **AWS EC2** servers, after rewriting **41** unit tests and refactoring the app to use **Docker**, as well as the latest LTI 1.3 standard, resolving **over a dozen** open GitHub issues.
- QA tested** and **rewrote** numerous widget components from **PHP to Django** in a team-wide port of our flagship product, **Materia**, which offers **25+** educational widgets to gamify and enhance a student's learning experience.

Hackathon Organizer, Developer

July 2025 – Present

Knight Hacks

Orlando, FL

- Organized the **Knight Hacks VIII** hackathon with **90% positive ratings**, as well as developing for our hacker management infrastructure, **Blade**, which achieved a peak of about **850** concurrent users during the event.
- Authored **attendance tracking systems** and an accompanying management interface in **React** and **tRPC/Node**, storing valuable metrics for **60** workshop/sponsor events and **6,400+** check-in instances throughout the weekend while implementing **rapid hot-fixes**.
- Designed new interfaces** for hacker-facing dashboards and leaderboards to drive engagement, as well as enhancing the staff-side check-in interface to allow **high volumes of check-ins**.
- Crafted **React Email** templates seen by **1,000+** applicants, communicating acceptance status in manner respectful to users.

Lead Developer, Vice President

May 2025 – Present

DenTraceX

Orlando, FL

- Co-founded a non-profit organization** dedicated to collecting dental records of missing persons, in order to cross-reference them with unidentified remains.
- Authored our organization infrastructure **entirely from scratch** using **React**, **ExpressJS/Node**, **PostgreSQL**, as well as our own self-hosted **AWS-compatible S3 bucket** storage solution.
- Constructed a **novel** code-based representation for jaw/tooth records, allowing easy searching/filtering to identify **real-world** human remains, intended for use by medical examiners and independent investigators.
- Working with a team of diverse disciplines to process and handle **sensitive medical information** and **maintain legal compliance**.

PROJECTS

VL-ADK | *Google ADK, Python, NVIDIA Nemotron, YOLO-E, React, FastAPI*

September 2025

- Created at **ShellHacks 2025**, won **1st place** for **NVIDIA's Hack the Future Challenge**, **2nd place** for **Microsoft's AI for Good Challenge**, and **3rd place** for **Waymo's Reimagining Navigation Challenge**.
- Mimicks the functionality of a **generalized VLA robotics model** using a fleet of **Google ADK** agents, powered by **NVIDIA Nemotron**.
- Allows **dynamic prompting** of a cost-effective **NVIDIA Jetson Orin**-powered robot, achieving **intelligent movement and spatial reasoning**, even with extremely limited sensing capabilities.
- With the assistance of a **YOLO-E** model for object detection, accomplished **structured paths** and **object search/navigation** behaviors, completely operated by **autonomous AI agents**.

Voiceboard AI | *React, TypeScript, Gemini API, Mermaid*

September 2024

- Created at **ShellHacks 2024**, won **1st place** for **Google's Hack for Social Good Challenge**.
- Authored the whiteboarding interface, allowing the entire app to be voice-controlled through **speech recognition** and **Google's Gemini AI** to massively improve accessibility.
- Implements prompt engineering through the **Gemini API** to create **context aware Mermaid diagram code**.

Lootcode | *React, Clerk, TypeScript, MySQL, Zx Shell, Docker, tRPC*

February 2024 – May 2024

- Created for **Knight Hacks Project Launch 2024**, won **1st place**.
- Developed a **gamified study application** for Data Structures and Algorithms, featuring **100+ unique problems**.
- Live-generates **Docker containers** to self-host code grading instances for *each user*, maintaining **server security**.
- Accumulated **50,000+** page visits and **500+** monthly active users, featured in **Linux Magazine's** OSS section.

TECHNICAL SKILLS

Languages: Java, Python, C/C++/C#, Lua, SQL, TypeScript, HTML/CSS, PHP

Software/Services: MySQL, PostgreSQL, MongoDB, Redis, Node.js, Google Gemini, NVIDIA Nemotron, Google ADK

Libraries/Frameworks: React, ExpressJS, FastAPI, Flask, Django, S3, Prisma, Clerk, JWT, tRPC, Tailwind CSS, Websockets, OpenCV, MediaPipe, pandas, YOLO-E

Developer Tools: Linux, Coolify, Git, Docker, VS Code, Eclipse, Cursor AI